

Brain dissection photogrammetry: a tool for studying human white matter connections integrating ex vivo and in vivo multimodal datasets

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Understanding the architecture of the white matter of the human brain is central to neuroscience and clinics. Despite major advances in tractography and white matter dissection, integrating these complementary techniques remains a longstanding challenge. Here, we introduce BraDiPho (Brain Dissection Photogrammetry), an open-access resource of high-resolution, fully textured 3D digital models of layer-by-layer white matter microdissection. The models are registered to their radiological space, allowing direct alignment of dissection and neuroimaging data. BraDiPho includes eight hemispheres, enriched with sample tractography bundles, anatomical annotations, and cortical atlases, establishing a unified framework for multimodal analyses. Four case studies demonstrate how BraDiPho supports anatomically grounded investigations, moving beyond classical side-by-side comparisons toward accurate integration of ex vivo dissection and in vivo tractography. All data, tools, and scripts are openly available, enabling customized research and educational applications. BraDiPho offers a framework to support multimodal investigations of human brain connectivity in both research and educational contexts.

A paramount contemporary challenge in neuroscience lies in unraveling the intricate network of white matter (WM) connections of the human brain¹. As conduits for electrical signals, their architecture regulates the flow of information throughout the brain, shaping

communication between cortical areas, determining the impact of lesions, and suggesting potential alternative routes for communication. Charting the organization of WM pathways is crucial for understanding the emergence of human behavior¹, the processes affected by

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their disruption, and the post-lesional recovery potential of specific cognitive functions². A reliable characterization of WM pathways is therefore essential for the neuroscientific community to explain the physiological interactions between and within cortical networks, and for the medical community (e.g., neurological, neurosurgical, neuroradiological, and neuro-oncological disciplines) to interpret the evolution of pathological and plastic changes occurring in this complex biological system. Multimodality is key for obtaining sound anatomical descriptions, and the integration of different sources of information requires a shared anatomically grounded space.

Diffusion magnetic resonance imaging (dMRI)-derived tractography has proven itself as an essential tool for the in vivo mapping of WM fiber pathways at the macroscale, providing accessible proxies of brain connections³. Concurrently, the cortex-sparing Klingler microdissection method has revived and refined classical WM anatomical studies, offering direct visualization of the WM architecture⁴. These techniques serve complementary purposes. In vivo tractography enables large-scale mapping of brain connectivity and supports clinical applications through patient-tailored reconstructions. Ex vivo microdissection remains the only anatomically grounded approach for assessing the reliability of tractography-derived representations⁵. Despite their synergy, their integration remains fragmented. Currently, anatomical data from microdissection and tractography are manually combined, often by sketching tractography-derived outlines onto dissection photographs^{6–11}. This juxtaposition exposes an inherent methodological gap: whereas tractography provides interactive 3D representations, microdissection is constrained to 2D images. Consequently, anatomical studies remain limited to qualitative descriptions from fixed perspectives, and the high-effort process of sourcing, preparing, and dissecting human specimens is not fully leveraged for broader scientific applications.

The lack of a structured method for directly integrating in vivo tractography and ex vivo microdissection confines the multimodal study of connectional neuroanatomy to a niche approach prone to approximations. Addressing this challenge requires sophisticated computational solutions capable of translating multimodal data across different spatial domains. In addition, broadening access to high-quality, high-resolution microdissection data in an interactive digital format is essential for refining our understanding of the architecture of WM pathways and assessing the limitations of in vivo tractography.

Here, we introduce BraDiPho (Brain Dissection Photogrammetry), a resource that provides high-resolution, fully digitized 3D models of ex vivo layer-by-layer WM microdissection. BraDiPho enables a reliable integration of in vivo structural and functional imaging with ex vivo microdissection, offering both qualitative and quantitative insights (Fig. 1). By bridging the gap between ex vivo microdissection and in vivo neuroimaging, BraDiPho enables more precise and accessible multimodal explorations of brain connectivity, laying a robust foundation for future subject-specific tractography validation.

Results

We report four key outcomes from the development and implementation of the BraDiPho resource. First, we present an open-access dataset of high-resolution digital 3D models capturing layer-by-layer ex vivo WM dissections. Second, we detail the integration of multiple modalities, including tractography, cortical parcellation atlases, and anatomical annotations, within these models, creating an interactive framework for multimodal neuroanatomical exploration (Fig. 1). Third, we demonstrate the versatility of BraDiPho through four distinct case studies, highlighting its capacity to support anatomical analyses ranging from spatially grounded assessments of tractography bundles to the direct fusion of dissection and tractography

Overview of BraDiPho resources provided with each of the eight ex-vivo 3D models currently available.

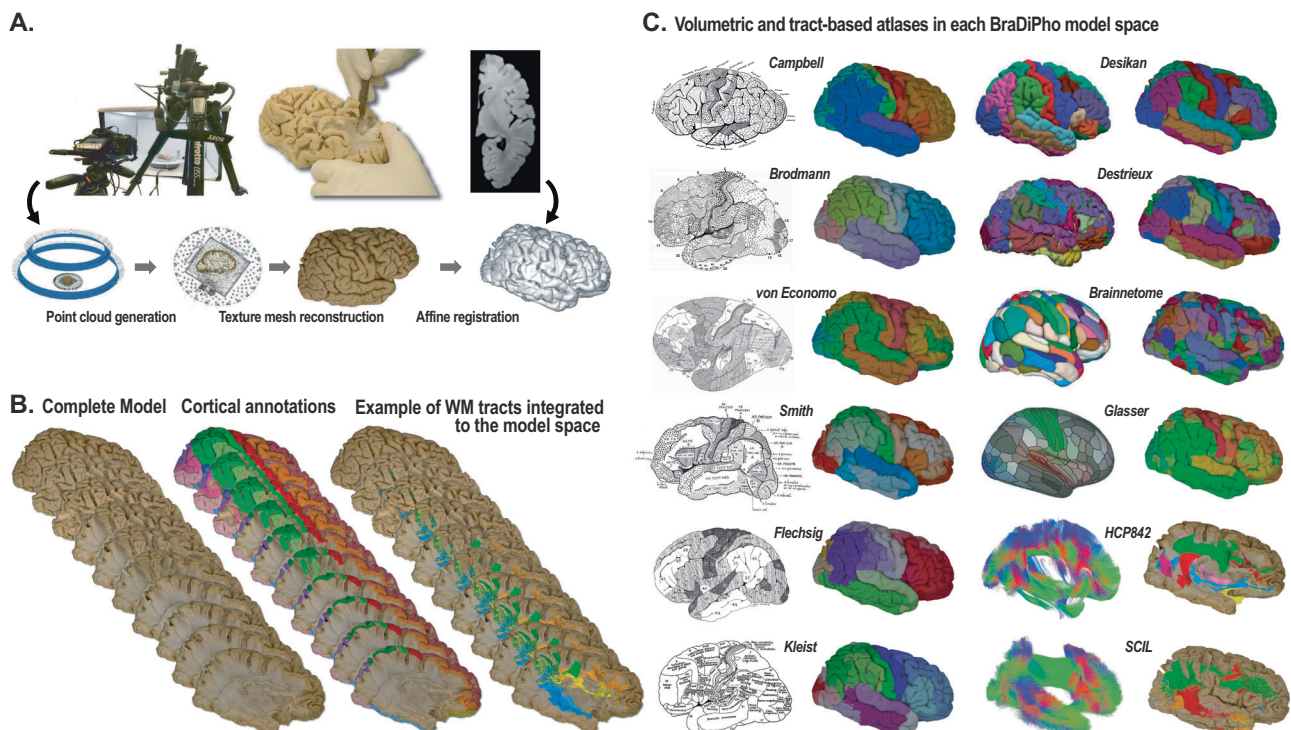


Fig. 1 | Overview of the resources provided for each of the eight ex vivo 3D models in BraDiPho. **A** Workflow of the photogrammetric reconstruction of cortex-sparing Klingler microdissection. Each dissection epoch is documented with 480 photographs acquired at 42-megapixel resolution over 360°. Textured meshes are reconstructed for each epoch and aligned with the surface model of the T1-MRI

of the specimen. For each BraDiPho model, users have access to all dissection epochs (**B**, left), cortical gyral annotations (**B**, middle), sample tractography bundles from an in vivo subject (**B**, right), and twelve reference atlases, including cortical parcellations and tractography atlases (**C**), registered to the radiological space of each specimen.

data (Figs. 2–5). Fourth, we provide a comprehensive suite of tools, scripts, computing environments, and online services to facilitate the development of custom studies based on external datasets.

The creation of a resource for the integrated exploration of ex vivo dissection and in vivo neuroimaging primarily requires: (i) the acquisition of high-resolution, anatomically precise images that comprehensively capture the entire dissection scene, (ii) the reconstruction of these images into cohesive 3D digital models, and (iii) the registration of the models into a standard reference space, capable of hosting neuroimaging data. To achieve this, T1-MRI acquisition was conducted on eight postmortem human hemispheres, and layer-by-layer microdissection was performed by two expert anatomists over 100 h. Across all specimens, we dissected 74 anatomical layers, henceforth referred to as epochs, progressively exposing major association pathways of the human brain. Including the intact convexity of each specimen ($N=8$), this yielded a total of 82 epochs. Each epoch was photographed across a full 360° range, resulting in 480 high-resolution images (42 megapixels) per epoch¹². This process, spanning 50 h of acquisition, generated 39840 images, for a total of ~2 TB of storage. Images capturing each epoch were independently processed to generate 3D digital models through photogrammetry¹³, a computationally intensive procedure requiring up to 2075 h, and producing ~100 GB of data derivatives. The resulting 3D dissection models were then recursively aligned to the T1-MRI scan of each specimen, enabling an interactive, multi-perspective exploration of WM architecture within a single multi-layered object (Fig. 1A). This approach fundamentally redefines traditional microdissection, overcoming the inherent limitation related to its static 2D rendering. With BraDiPho, digitized dissections can be dynamically explored across multiple perspectives, preserving the anatomical continuity of fiber contingents and their spatial relationships. Moreover, the integration of epochs into a single multi-layered object preserves the temporal dimension of dissection, allowing users to navigate through successive layers by rewinding the stepwise exposure of WM pathways. BraDiPho also supports the incorporation of neuroimaging data from multiple modalities, establishing a robust multimodal framework for studying WM anatomy.

Beyond anatomical digitization, BraDiPho embeds a selection of cortical and subcortical imaging datasets registered to each dissection model, alongside anatomy-driven custom annotations (Fig. 1B, C). These complementary data concur in a multimodal integrative framework for the study of WM anatomy. All datasets are freely available via the BraDiPho website (<https://bradipho.eu/>), which also features an online registration suite for users to integrate their own tractography data into the dissection models. In addition, a series of offline scripts supporting customized analyses are available on GitHub (<https://github.com/minilabus/bradiphophy/tree/main/scripts>).

Datasets overview

BraDiPho comprises a growing collection of 3D digital photogrammetric models of layer-by-layer Klingler microdissection. The dataset currently includes eight hemispheres, four left (spc-01, spc-04, spc-16, spc-18) and four right (spc-02, spc-10, spc-17, spc-19). Each model documents the progression of the stepwise cortex-sparing Klingler microdissection process, spanning 7 to 12 distinct epochs. Specimens 01, 02, 04, 10, 16, and 19 were dissected using a latero-medial approach, first revealing short posterior transverse connections before progressively isolating deeper and longer fibers of both dorsal and ventral longitudinal systems. While specimens 01, 02, 04, and 10 are primarily dedicated to dorsal association fibers, specimens 16, 17, 18, and 19 focus on the dissection of the ventral system of longitudinal WM connections. Notably, the dissection of specimens 17 and 18 begins with the exposure of basal connections of the temporal lobe, followed by the progressive identification of ventral longitudinal connections of the frontal cortex.

Each specimen is supplemented with manual annotations delineating gyral landmarks, originally traced on the intact convexity, then automatically projected across successive epochs. Similarly, widely used cortical parcellation atlases derived from multiple modalities, such as cytoarchitectonic, myeloarchitectonic, and functional mapping, are mapped onto the cortical surface of each specimen and transferred along all the dissection epochs (Fig. 1C). The included atlases are: Brainnetome atlas¹⁴, Brodmann atlas^{15,16}, Campbell atlas¹⁷, Economo atlas¹⁸, Flechsig atlas¹⁹, Desikan-Killiany atlas²⁰, Destrieux atlas²¹, Glasser atlas²², Kleist atlas²³ and Smith atlas²⁴.

A curated set of tractography bundles registered into the space of each dissection model has been selected from two well-recognized atlases, namely the HCP842²⁵ and the SCIL atlas²⁶ (Fig. 1C). The selection includes: the arcuate fasciculus (AF), the acoustic radiation, the cingulum, the frontal aslant tract, the inferior fronto-occipital fasciculus (IFOF), the inferior longitudinal fasciculus, the middle longitudinal fasciculus, the optic radiation, the pyramidal tract, the superior longitudinal fasciculus, the uncinate fasciculus and the vertical occipital fasciculus.

In addition, BraDiPho includes a custom dataset of tractography bundles aligned with the dissection models and representing the associational connectivity of the angular gyrus (AG)¹¹. This dataset, along with corresponding manual WM annotations for spc-01 and spc-02, is available for download and further included in the online viewer featured by the Preview section of the website (<https://bradipho.eu/2-3d-visualizer-r.html>). While the online viewer provides a lower-resolution version of what can be achieved with the offline browsing of the data, it demonstrates the potential application of the BraDiPho framework for custom anatomical analyses.

Custom experience

All materials provided within BraDiPho converge into an integrated framework for the multimodal analysis of human brain connectivity. The Download section of the website provides all the essential components for studying WM anatomy through a multimodal approach. In addition, this resource allows users to integrate their own neuroimaging data into the photogrammetric models of ex vivo dissection, enabling customized studies.

The Tool section of the website supports online linear registration of tractography bundles from their reference space to any available dissection model (<https://bradipho.eu/5-tool-convert.html>). The warp and affine matrices for non-linear registration from MNI space to dissection models are available at <https://zenodo.org/records/11192915>²⁷. A suite of custom scripts hosted on GitHub (<https://github.com/minilabus/bradiphophy/tree/main/scripts>)²⁸ provides tools for converting volumetric data into surface representations and supporting the interaction with meshes and point clouds. To ensure reproducibility, we provide containerization instructions through a Docker file. Tutorials documenting data browsing, manual annotation, and integration of dissection and tractography data are available in the Preview section of the website (<https://bradipho.eu/2-3d-visualizer-r.html>).

Applicability examples

This section showcases four case studies that illustrate possible practical applications of BraDiPho. Each case demonstrates how integrating ex vivo dissection and in vivo tractography in the same radiological space enables both direct comparison and combined anatomical analyses across modalities.

Case 1 - Integrating dissection and tractography for white matter analysis. Digitizing the different stages of ex vivo dissection into interactive and multi-layered 3D models broadens the impact of a single session of dissection. Indeed, BraDiPho allows to rewind the dissection procedure along different time points and to visualize anatomical information from multiple perspectives, thus enabling

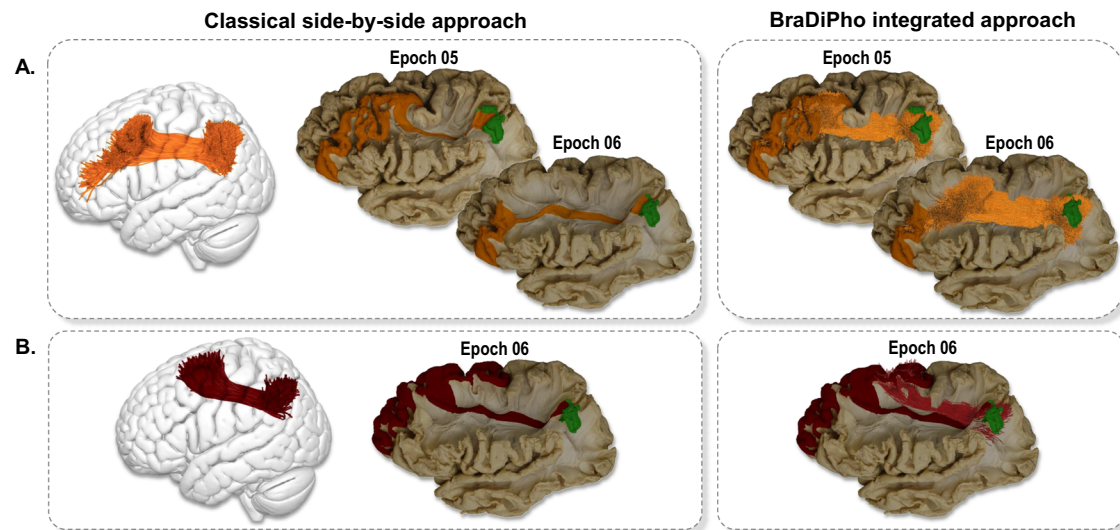


Fig. 2 | Direct integration of tractography and dissection with BraDiPho. Evaluation of the anatomical extent of (A) AG-MFG and (B) AG-SFG connections as reconstructed with in vivo tractography in 39 healthy participants and revealed by ex vivo microdissection of spc-02. The first column illustrates the standard side-by-side comparison between tractography and dissection. The second column shows their direct integration as implemented in BraDiPho. In (A), tractography and

dissection evidence of AG-MFG connectivity converge, uncovering fundamental principles of brain organization. In (B), apparent agreement between modalities in the classical side-by-side view is challenged by the integrated BraDiPho approach, which reveals discrepancies in the reconstruction of AG-SFG connections achieved with tractography and dissection. AG angular gyrus, MFG middle frontal gyrus, SFG superior frontal gyrus, spc specimen.

post-hoc investigations and repurposing data for diverse applications. Concurrently, integrating dissection models into a defined radiological space bridges the traditional gap between dissection and neuroimaging studies, allowing for direct cross-evaluation of findings.

As showcased in this work focusing on WM dissection and tractography, BraDiPho marks a significant paradigm shift in the evaluation of in vivo neuroimaging data with information coming from anatomical studies. Case 1 illustrates the study of the connectivity between the AG and both the middle (MFG) and superior (SFG) frontal gyri, leveraging the integration of tractography and dissection evidence. Tractography data were sourced from the concatenation of 39 normative individual tractograms and were non-linearly registered in the ex vivo dissection model of specimen 02. Considering AG-MFG connections (Fig. 2A), the analysis of the dissection model reveals a hierarchical, two-tiered organization: shorter, lateral fibers connecting the posterior portion of the MFG are exposed in an earlier dissection epoch (epoch 05), whereas longer fibers extending more anteriorly in the MFG are revealed in a deeper dissection epoch (epoch 06). The integrated exploration of in vivo tractography and ex vivo dissection demonstrates convergent findings. Indeed, when tractography is aligned with the dissection models, the most lateral dissection layer reveals only shorter, superficial fibers of the tractography bundle, while longer ones remain masked. As the dissection progresses to deeper layers, longer streamlines become visible, reflecting the same fiber organization revealed through dissection. These observations reaffirm a longstanding principle of WM organization first described by Meynert²⁹, which states that short fibers typically course more superficially than longer ones.

While this instance demonstrates a strong agreement between ex vivo dissection and in vivo tractography data, Fig. 2B exposes the limitations of the conventional side-by-side approach, commonly regarded as the current gold standard for the comparison of the two modalities³⁰. While traditional side-by-side comparison of AG-SFG connections demonstrates an agreement between tractography and dissection findings (Fig. 2B, “Classical side-by-side approach”), their direct integration as implemented in BraDiPho highlights a mismatch (Fig. 2B, “BraDiPho integrated approach”). Indeed, the tractographic reconstruction courses more dorsally and terminates more posteriorly

in SFG compared to the dissection evidence. This mismatch stresses the bidirectional nature of interpretations enabled by BraDiPho, where discrepancies invite consideration of both tractography limitations and dissection constraints. Interpretations must account for the nature of the tractography reconstruction and its spatial relationship with the dissection evidence. Here, the population-based tractography, derived from 39 individuals, fails to capture a connection evident in the dissection of a single specimen, suggesting a potential bias in the tractography data. However, this does not invalidate the streamlines of the tractography reconstruction as anatomically implausible. Since they course below the dorsal PrCG, which remains intact at this stage of dissection, the connection may simply be obscured rather than absent. More generally, while tractography is prone to artefactual reconstructions owing to its reliance on indirect measurements, dissection is not susceptible to such artifacts, and what is revealed through dissection unequivocally exists. Yet, dissection is not exhaustive, as the visibility of fiber pathways depends on the specific anatomical layers that have been exposed at a given stage. The interpretation of mismatches, therefore requires users to read the anatomical context provided by the dissection models and, at the same time, to be knowledgeable of the technical features of the tractographic reconstruction under evaluation. The example at issue further underscores the value of retaining individual dissection data even when evaluating population-based tractography reconstructions, as multiple one-to-one comparisons are essential for identifying anatomical features that would be lost in group-averaged representations.

Case 2 - Characterizing dissection with tractography. By providing a shared anatomical space for the integration of in vivo tractography and ex vivo dissection, BraDiPho supports the incorporation of information coming from one modality into the other. Case 2 shows how tractography data can be directly mapped onto the texture of the dissection models. Figure 3 illustrates the evolution of the logarithmic signed distance between a sample bundle, the AF from the SCIL atlas, and the dissection model of specimen 01 across different epochs, documenting the assessment of how the anatomical trajectory and the cortical terminations of a bundle align across the two modalities on sequential epochs. In panel A, the signed distances projected on epoch

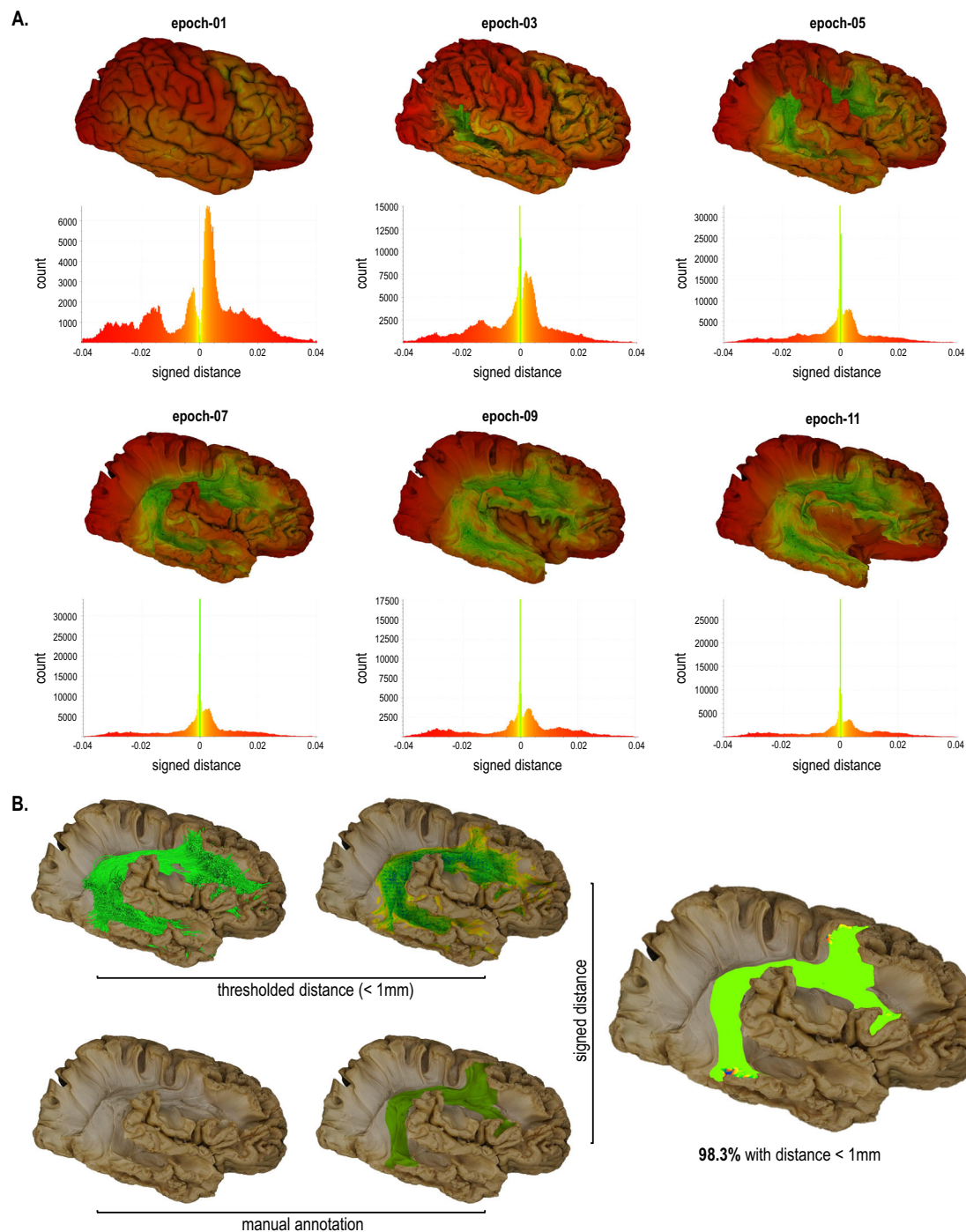


Fig. 3 | Quantitative evaluation of the spatial correspondence between tractography-derived bundles and layer-by-layer white matter dissection.

Panel (A) displays the evolution of the signed distance between the arcuate fasciculus (AF) from the SCIL atlas and dissection models of spc-01 across multiple epochs. The photogrammetric model of each epoch displays distance values mapped onto its surface using a green–yellow–red color gradient (green = closer, red = farther from the bundle). Below each model, a histogram illustrates the distribution of signed distances between the bundle and the corresponding epoch. The progression illustrates how the tractography bundle increasingly overlaps with the dissection planes, with distances approaching zero, thus indicating greater anatomical overlap. By epoch 07, the closest match is observed, highlighting the best spatial correspondence between tractography and dissection evidence. Panel (B) illustrates the spatial correspondence between the

tractography bundle and the manual annotation of that same connection on epoch 07. The distance between the tractography bundle and the dissection epoch is filtered with a distance threshold of 1 mm to identify areas of close spatial proximity. The corresponding WM fibers in the dissection are manually annotated based on visual inspection of the same model. To evaluate spatial correspondence between the two, the signed distance between the annotated fibers and the thresholded tractography overlap is computed and projected onto the annotation as a scalar field. This field is visualized using a green–yellow–red color scale, where green indicates the closest proximity. A quantitative measure is then derived by calculating the proportion of annotated points falling within a 1 mm distance of the tractography overlap. In this example, 98.3% of the annotated AF fibers fall within this range, supporting a strong spatial match between tractography and dissection evidence at this stage. spc specimen.

O1 highlight the termination territories of the AF in orange, marking areas of the bundle laying closest to the dissection model at this stage. As dissection progresses, the tractographic reconstruction increasingly intersects with the dissection planes, with a higher number of points of the tractography reconstruction showing distances close to zero, indicating greater anatomical overlap. By epoch 07, the tractography bundle exhibits the best correspondence with the dissection model, with the highest number of points showing near-zero distance values, denoting optimal spatial overlap between the dissection epoch and tractography reconstruction.

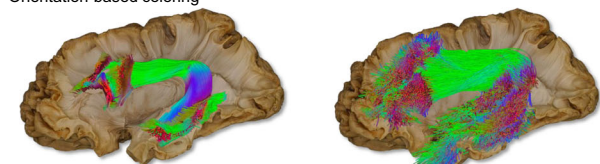
Panel B further refines this analysis by overlaying the smallest distance interval between the tractography bundle and epoch 07 onto the dissection model, considering a 1 mm distance threshold. This visualization highlights areas where tractography and dissection most closely intersect, offering a direct representation of the anatomical course and cortical terminations of the reconstructed tractography bundle on the texture of the dissection model. This approach serves as a heuristic tool to guide anatomical exploration and identify candidate dissection epochs that may warrant closer inspection for the presence of a tractography-reconstructed connection of interest. To provide a quantitative measure of consistency between the tractographic reconstruction and the corresponding fibers as identified on a specific dissection epoch, we implemented an additional proximity analysis leveraging manual WM annotations. Specifically, we annotated dissected fibers corresponding to the AF on the photogrammetric model of epoch 07 and calculated the signed distance between this annotation and the previously computed thresholded overlay of the tractography bundle on the dissection epoch. It generated a scalar field projected on the annotated fibers, visualized using a green–yellow–red gradient, where green denotes areas of closest correspondence with the tractography projection. By calculating the proportion of points of the annotation falling within a <1 mm distance threshold, we obtain a quantitative

estimate of overlap between the tractography reconstruction and the dissected fiber trajectory for a specific epoch. In the case demonstrated in Fig. 3, 98.3% of points of the annotated fiber pathway fall within 1 mm distance from the overlap map of the tractography reconstruction and the dissection epoch. Unlike traditional side-by-side comparison approaches leveraging the manual outlining of the tractography representation on 2D dissection images, this projection provides a direct and anatomically grounded comparison, refining how we assess the reliability of tractographic reconstructions.

Case 3 - Characterizing tractography with dissection-based cortical annotations. While Case 2 demonstrates how in vivo tractography can be used to characterize ex vivo dissection models, Case 3 shows the reverse, directly mapping information derived from the dissection onto tractographic reconstructions. Figure 4 illustrates the analysis of the left AF and the right IFOF from both the HCP842 and SCIL tractography atlases based on their cortical terminations as defined by the gyral architecture of individual specimens (spc-02 and spc-16 for left AF and right IFOF, respectively). Cortical annotations manually delineated on the specimens were used to parcel each bundle based on its cortical termination sites. The cortical termination-based coloring scheme provided here (Fig. 4A, B, second row) displays the different gyral territories associated with each bundle, providing an anatomical context that is absent in traditional orientation-based tractography coloring (Fig. 4A, B, first row). In the case of the left AF of the HCP842 atlas (Fig. 4A, second row), anterior terminations predominantly occur in the inferior frontal gyrus (IFG, gold), followed by a similar proportion in the MFG (orange) and the precentral gyrus (PrCG, red). In contrast, the left AF from the SCIL atlas primarily projects to the middle and inferior frontal gyri, with minimal involvement of the PrCG. Similarly, the right IFOF (Fig. 4B) exhibits strong posterior terminations in the middle occipital gyrus (MOG, pink) and the lingual gyrus (LG, light blue, not visible in the

A. Arcuate Fascicle (AF) on specimen 02

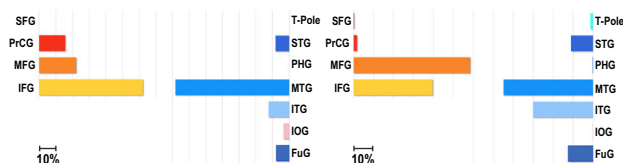
Orientation-based coloring



Cortical termination-based coloring

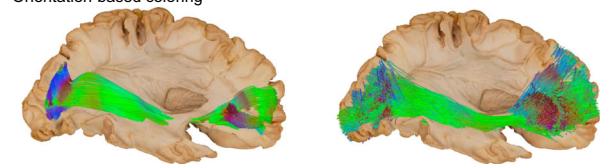
HCP842

SCIL



B. Inferior Fronto-Occipital Fascicle (IFOF) on specimen 16

Orientation-based coloring



Cortical termination-based coloring

HCP842

SCIL

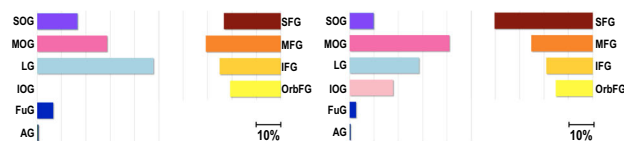


Fig. 4 | Integration of dissection-derived cortical annotations with tractography. Manual gyral annotations from BraDiPho specimens were used to guide the parcellation of homologous fiber bundles from the HCP842 and SCIL tractography atlases. Panel (A) illustrates the arcuate fasciculus (AF) in spc-02, while panel (B) presents the inferior fronto-occipital fasciculus (IFOF) in spc-16. Each bundle is shown with two coloring schemes: orientation-based (top row), where streamlines are colored according to their local orientation, and cortical termination-based (bottom row). Bar plots below each reconstruction display the proportion of streamlines terminating in different cortical gyri. For the IFOF, a substantial

proportion of streamlines terminate in the lingual gyrus (LG, light blue), though these are not visible due to their medial projection within the 3D dissection model. Cortical annotation names and color codes are provided in Supplementary Data 1. AG angular gyrus, FuG fusiform gyrus, IFG inferior frontal gyrus, IOG inferior occipital gyrus, ITG inferior temporal gyrus, LG lingual gyrus, MFG middle frontal gyrus, MOG middle occipital gyrus, MTG middle temporal gyrus, OrbFG fronto-orbital gyrus, PHG parahippocampal gyrus, PrCG precentral gyrus, SFG superior frontal gyrus, SOG superior occipital gyrus, spc specimen, STG superior temporal gyrus, T-pole temporal pole.

figure), followed by the superior occipital gyrus (SOG, purple) for both tractography atlases. However, as opposed to the HCP842 atlas, the right IFOF of the SCIL atlas also extends to the inferior occipital gyrus (IOG, light pink). These qualitative observations are quantitatively supported by the termination distribution analyses displayed in Fig. 4, bottom.

Case 4 - Characterizing tractography with atlas-derived cortical annotations. Leveraging the multimodal datasets included in BraDiPho, this labeling approach can be applied across multiple brain parcellation schemes, enabling comparative analyses. Figure 5 showcases ten distinct cortical atlases, projected onto specimen 01, contextualizing both AF and IFOF terminations. These atlases encompass classical cyto- and myelo-architectonic frameworks (e.g., Brodmann, Campbell, Von Economo, Flechsig, Kleist, Smith), as well as contemporary connectivity-based parcellations (e.g., Glasser, Brainnetome, Desikan, Destrieux). This comparative perspective reveals that historical atlases, such as those of Brodmann and Von Economo, provide broader cortical segmentations, whereas connectivity-based atlases like Brainnetome and Glasser offer finer-grained parcellations that better capture fiber termination heterogeneity. By integrating these parcellations within BraDiPho's anatomical framework, we directly assess how well different cortical maps align with the individual anatomical features observed in ex vivo dissections.

This alignment facilitates a biologically grounded evaluation of tractography-based fiber attributions, ensuring that variations in cortical labeling are interpreted within a realistic anatomical context. By combining multiple parcellation schemes in combination with direct anatomical observations, BraDiPho enhances the anatomical and functional interpretation of structural connectivity, refining our understanding of how WM pathways map into different definitions of cortical organization.

Discussion

Connectional neuroanatomy has become central to neuroscience, as WM fibers are increasingly recognized for their role in cognition,

behavior, and brain pathologies¹. Over the past two decades, dMRI-based tractography and innovative adaptations of Klingler's dissection have driven major advances in the mapping of WM connections. However, despite their respective and complementary strengths, the lack of effective integration between them limits a comprehensive and anatomically reliable characterization of the wiring of the brain. Tractography provides non-invasive, in vivo reconstructions of WM pathways, but remains affected by methodological limitations such as false positives and poor anatomical specificity⁵. Conversely, ex vivo dissection offers direct anatomical evidence, but is inherently destructive and limited in spatial resolution and dimensionality, as it is typically confined to 2D photographs. The integration of these approaches remains a longstanding challenge, as previous studies have typically considered them in isolation, resulting in fragmented, modality-specific insights. BraDiPho bridges this gap by converting traditional ex vivo dissections into high-resolution, fully textured 3D digital models registered to a standardized radiological space. This enables direct spatial alignment and multimodal integration with neuroimaging data, establishing a unified framework for WM exploration.

The first major contribution of BraDiPho lies in its collection of digital microdissection models, encompassing eight human hemispheres, each dissected over 7 to 12 progressive epochs. Klingler's layer-by-layer microdissection requires considerable expertise in both preparation and dissection of the specimens, limiting its practice to a small number of specialized laboratories worldwide³¹. As a result, researchers, clinicians, and students often rely on indirect representations of WM anatomy, with restricted access to direct anatomical evidence. BraDiPho addresses this limitation by offering open access to the largest and most anatomically detailed dataset of its kind. Crucially, the digitization of the full dissection process into interactive 3D models further allows a single dissection session to be repurposed for multiple scopes, significantly expanding its impact. Users can navigate through dissection epochs, change viewpoints, and explore regions of interest according to their specific research or educational goals.

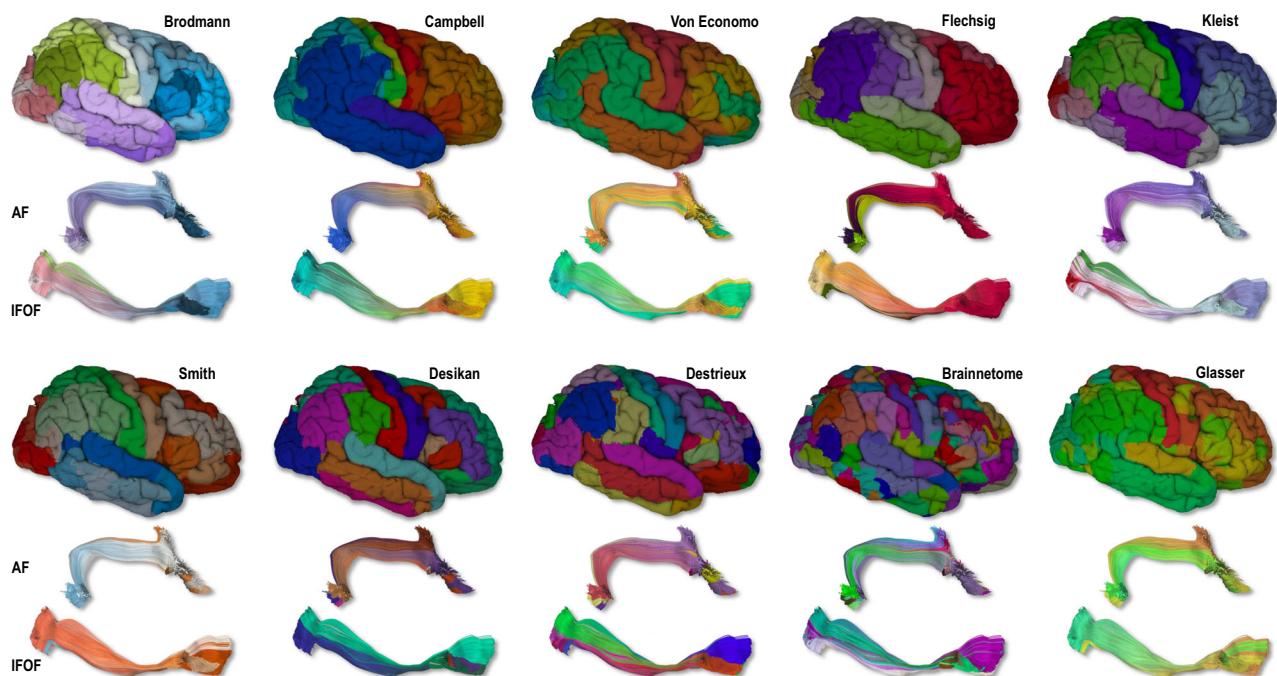


Fig. 5 | Cortical labeling of major association bundles according to ten cortical parcellation schemes. The arcuate fasciculus (AF) and inferior fronto-occipital fasciculus (IFOF) from the HCP842 atlas are shown with cortical termination-based coloring under ten different cortical atlases. These include classical cyto- and

myeloarchitectonic maps (Brodmann, Campbell, Von Economo, Flechsig, Kleist, Smith) as well as modern connectivity-based parcellations derived from functional and structural data (Desikan, Destrieux, Brainnetome, Glasser). Cortical annotation names and color codes are provided in Supplementary Data 1.

Unlike traditional 2D photographs, which fix the anatomy to a single angle, BraDiPho allows dynamic exploration, whether for studying the course of WM fibers and their relationship with other anatomical structures, planning neurosurgical approaches, or teaching neuroanatomy.

Beyond improving accessibility, BraDiPho redefines the anatomical evaluation of tractography reconstructions by grounding it in *ex vivo* dissection evidence, offering a transformative framework to tackle one of the major challenges in connectional neuroanatomy. Indeed, the reliability of dMRI-based tractography has long been questioned due to the lack of a ground truth at the macroscale³². As demonstrated in Case 1 of the Results section, traditional side-by-side comparisons between tractography and dissection data can create a misleading impression of agreement, as visual similarities alone do not confirm anatomical accuracy. By supporting spatial alignment between the two modalities, BraDiPho enables a direct one-to-one assessment of tractography reconstructions against anatomical ground truth. This allows researchers to systematically detect and correct for biases, artifacts, and false positives introduced by tractography algorithms³³, offering an anatomically grounded framework that improves the interpretability and accuracy of *in vivo* tractography findings.

BraDiPho also enables the integration of information across modalities, allowing each technique to inform the other. As exemplified in Case 2, tractography can be directly mapped onto the dissection texture, revealing how bundle trajectories align with anatomical landmarks and furthering joint anatomical analyses on the extent of WM connections. Conversely, Case 3 documents the integration of information about the unique gyral folding patterns of the specimen into the tractography reconstruction, demonstrating how anatomical features can contextualize tractography findings. These use cases illustrate just a few possible ways of integrating information from dissection and tractography, showing that BraDiPho enables a bidirectional exchange where derivatives of each modality characterize and enrich the other.

Further expanding this multimodal integration, Case 4 introduces the contribution of other neuroimaging data included in BraDiPho, namely, cortical parcellation schemes from multiple modalities. By registering these atlases in the radiological space of each specimen and projecting them onto the different dissection layers, BraDiPho extends tools traditionally used in neuroimaging analysis to the realm of *ex vivo* dissection. Integrating population-derived atlases with individual dissection data enables the application of widely used neuroimaging tools to contextualize accurate anatomical representations, supporting multimodal comparisons and underscoring the relevance of individual variability. While BraDiPho provides sample complementary ready-to-use datasets, it also offers the methodological framework and scripts used for their generation. Indeed, the contribution of BraDiPho extends beyond its datasets, providing all necessary tools and scripts for users to integrate their own imaging data with the dissection models. By supporting this integration, BraDiPho not only advances normative neuroanatomical research but also holds strong potential for clinical applications, including neurosurgical planning and education. Our approach aligns well with recent clinical frameworks such as the “à la carte connectome” proposed by Valdes and colleagues, in which neurosurgical cases are mapped onto MNI space and combined with atlas-based tractography to estimate the involvement of WM pathways and potential functional risks³⁴. While this approach provides a high-level overview of structural connectivity, BraDiPho complements it by enhancing anatomical reliability and providing an immersive, interactive environment. This expands the possibilities for virtual neurosurgical training, allowing the anatomical complexity of WM to be explored within an anatomically grounded 3D space, supporting individualized neurosurgical training and case planning.

The current release of BraDiPho focuses on major longitudinal and transverse association pathways of the human brain. Designed as an open-ended resource, it will continue to expand with the inclusion of an increasing number of dissection models for the refinement of associational connectivity and dedicated to projection and commissural fibers. Due to the progressive tissue removal inherent in Klingler’s dissection, full characterization of all fiber pathways within a single specimen is not feasible. Therefore, our approach relies on collecting dissections from multiple individuals to be included in BraDiPho. This growing dataset will not only broaden the applicability of dissection data to different studies, it will also provide an account of interindividual variability. Indeed, although individual specimens cannot capture the full range of interindividual variability when considered singularly, preserving their unique anatomical features allows for the identification of variants and recurring patterns across specimens. By aligning dissection models with their T1-MRI and thus allowing registration of specimen-derived annotations to a standardized space, BraDiPho lays the groundwork for future group-level analyses across multiple hemispheres. This scalability supports broader anatomical insights while retaining individual specificity. Looking ahead, the most critical step towards effective tractography validation will be the integration of dMRI-based tractography and *ex vivo* microdissection performed on the same specimen. While the current resource levels up methodological differences across modalities, direct within-subject comparisons would further enhance anatomical accuracy and improve tractography validation efforts. This single-subject, multimodal approach would reduce variability introduced by inter-individual differences and holds strong potential for reciprocal methodological refinements: tractography can guide more targeted and tissue-sparing dissections, while dissection findings can inform tractography parameter optimization and anatomical constraints.

BraDiPho represents a major advancement in the study of the structural connectivity of the human brain at the macroscale, addressing long-standing challenges in anatomical characterizations, accessibility, and multimodal integration. By providing an open-access, high-resolution, and fully interactive dataset, it breaks barriers to accessing WM microdissection data, ensures a higher level of accuracy in interpreting tractography findings, and introduces a framework for integrating *ex vivo* and *in vivo* neuroanatomical data. While Klingler’s dissection offers direct anatomical visualization of WM tracts, we also acknowledge its inherent limitations, particularly in resolving complex fiber configurations such as crossing, kissing, or branching fibers, and layered microstructures, which may be disrupted during the dissection process. Complementary techniques such as tracer-based methods -despite being invasive and limited to non-human models - provide high specificity and directional information³⁵, while polarized light imaging (PLI) allows for high-resolution, orientation-sensitive mapping of myelinated fibers, capturing angular and laminar details beyond the reach of dissection³⁶. Looking ahead, integrating dissection with *ex vivo* diffusion tractography, histological imaging, tracing, or PLI data with the BraDiPho framework could help overcome the limitations of individual modalities and offer a more comprehensive understanding of human WM architecture. As it continues to expand, BraDiPho has the potential to become a cornerstone resource for researchers, clinicians, and educators seeking a comprehensive and anatomically grounded approach to WM exploration.

Methods

BraDiPho currently includes eight human brain specimens: four right (01, 04, 16, and 18) and four left hemispheres (02, 10, 17, and 19), each dissected over 7 to 12 progressive epochs (Fig. 6A for right hemispheres, Fig. 6B for left hemispheres, first column for each specimen, and Table 1). All specimens were processed through the same methodological pipeline detailed below.

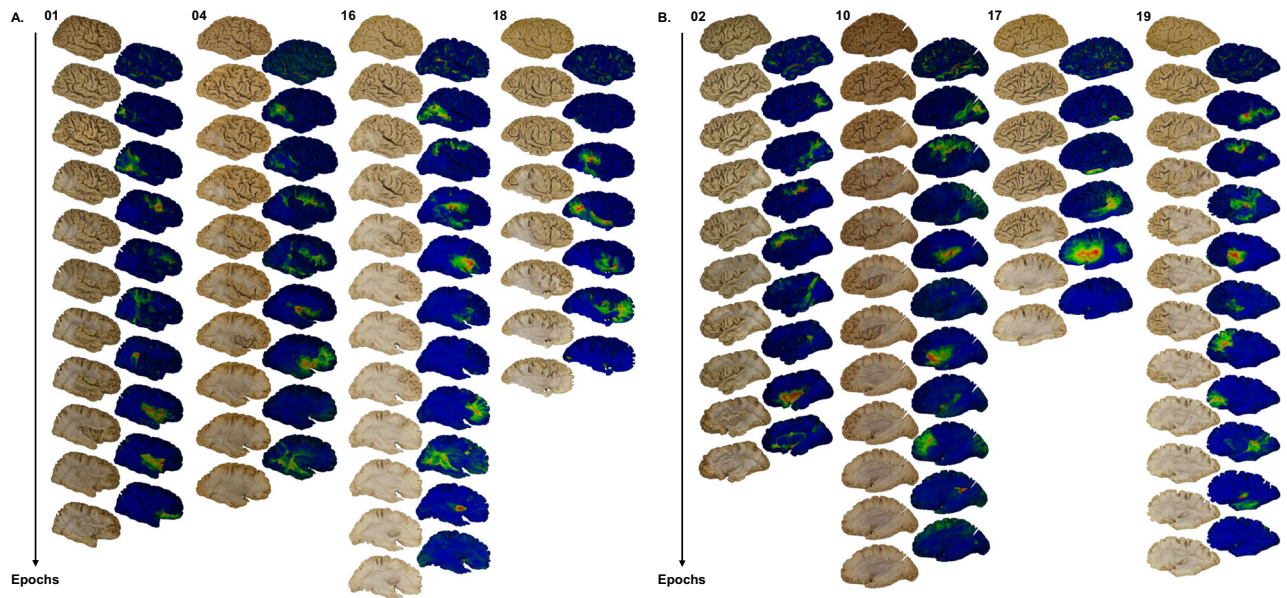


Fig. 6 | All dissection epochs of the eight hemispheres currently provided in BraDiPho. For each specimen, both right (A) and left (B) hemispheres, the first column displays the original photogrammetric models across all dissection epochs, from the intact convexity to deeper anatomical layers. The second column presents the scalar field encoding the absolute distance between each dissection epoch and

its immediate predecessor, rendered onto the surface texture of the later epoch. Blue regions denote minimal or no displacement, indicating anatomical stability, whereas green to red regions correspond to localized displacements, which systematically localize where tissue has been removed. This spatial pattern confirms the accuracy of the registration procedure.

Table 1 | Summary of the dissection features of each specimen currently included in BraDiPho

Specimen	Hemisphere	Epochs	Latero-medial dissection	Dorsal association fibers dissection	Ventral association fibers dissection	Basal association fibers dissection
spc-01	Right	11	✓	✓	–	–
spc-02	Left	10	✓	✓	–	–
spc-04	Right	10	✓	✓	–	–
spc-10	Left	12	✓	✓	–	–
spc-16	Right	12	✓	–	✓	–
spc-17	Left	7	–	–	✓	✓
spc-18	Right	8	–	–	✓	✓
spc-19	Left	12	✓	–	✓	–

Specimen preparation protocol

The postmortem human brain hemispheres included in BraDiPho originate from the Brain Anatomy Data Bank (BADaBank) of the Structural and Functional Connectivity Lab Project, Department of Anatomic-Pathology and Department of Neurosurgery of Santa Chiara Hospital (Azienda Provinciale per i Servizi Sanitari (APSS), Trento, Italy). The use of individual brain hemispheres sourced from routine autopsies carried out at Santa Chiara Hospital is regulated under national and institutional policies that, as stated and approved in the ethics protocol, authorize their use for scientific purposes without requiring prior consent. The study was approved by the Ethical Committee of APSS (N° 06/2018, renewed 02/2024) and complies with all relevant ethical regulations, including the regulation on the use of brain hemispheres sourced from autopsies carried out at Santa Chiara Hospital for scientific purposes. Reporting on sex, age, and gender is not relevant to this study as no influence of these factors is investigated. This information is further protected by the bylaw on privacy protection referenced in the study approval of the APSS Ethical Committee. Each hemisphere was prepared following an adapted version of Klingler’s original protocol³⁷. Specimens were fixed in a 10% formalin solution for 40 days, then frozen at – 80 °C for 30 days before being thawed. The arachnoid, pia mater, and vessels were meticulously removed during this process. The soaking, freezing, and subsequent

defrosting of specimens allow water molecules to permeate tissues, expanding their volume and forming ice crystals, separating WM fibers and facilitating their isolation during dissection³¹. Specimens were stored in formalin-filled jars until MRI acquisition and dissection.

T1 MRI acquisition and processing

MRI scans were acquired using a 1.5 Tesla GE scanner (GE Healthcare, Boston, Massachusetts, USA) with a T1-weighted volumetric sequence (1 mm slice thickness, 24 mm field of view, no gap acquisition). The jar containing the specimen was removed from the images through brain extraction. T1-MRI images were aligned to the anterior commissure–posterior commissure (AC–PC) plane of the Montreal Neurological Institute (MNI) template. A mesh model of each hemisphere was then generated from the multislice volumetric ex vivo T1-MRI using Monte Carlo methods and smoothing algorithms.

Layer-by-layer Klingler dissection

Dissections were performed using a refined, cortex-sparing version of the Klingler technique, proceeding layer by layer with wooden spatulas and following a latero-medial approach (Fig. 6A, B). The dissection of four hemispheres (left: 02, 10; right: 01, 04) focused on the posterior transverse (PTS) and superior longitudinal systems (SLS), whereas the other four hemispheres (left: 17, 19; right: 16, 18) were dedicated to

delineating the inferior longitudinal system (ILS)³⁸. Given the methodical nature of layer-by-layer dissection, involving the gradual removal of superficial layers to access deeper ones, in addition to isolating ILS fibers in the last four specimens, we successfully extracted the superficial layers of the SLS and PTS in two hemispheres (spc-16, spc-19), and the basal longitudinal system (BLS) in the remaining two (spc-17, spc-18). In all eight hemispheres, dissection commenced from the intact convexity, progressing with systematic decortication of the sulci while preserving the gray matter at the summit of the gyri.

The extraction of the PTS entails two distinct epochs. The initial epoch encompasses superficial fibers connecting the posterior sector of the middle temporal gyrus (MTG) to the inferior parietal lobule, alongside intralobar occipital fibers running parallel and just posterior to them. The second epoch includes longer and deeper fibers connecting the inferior temporal and superior parietal cortices.

The dissection of the SLS starts with the extraction of fronto-parietal fibers. A superficial epoch reveals connections between the AG and the posterior frontal cortices, while a deeper epoch comprises longer fibers extending to the anterior frontal cortices. Deep fronto-temporal fibers are exposed after the removal of the AG.

The first step in dissecting the ILS involves the identification of its stem, *i.e.*, the bottleneck region of the external/extreme capsule through which all ventral interlobar association fibers pass. This requires the removal of the opercula and the excision of the underlying insula. From the stem, the dissectionist follows the trajectory of WM fibers both anteriorly and posteriorly. Shorter fibers linking the inferior frontal to the anterior temporal cortices are located more lateral and closer to the cortical surface, while longer fibers extending to the temporal, occipital, and parietal cortices run more medially. Characterizing the entire wiring of the ILS typically requires 3 to 4 epochs.

Dissection of the BLS starts from the basal surface of the hemisphere. Analogous to previously described systems, the BLS features a superficial layer of short fibers and a deeper one of longer fibers.

These dissection guidelines were adapted on a case-by-case basis, as individual anatomy and preparation quality influence tract accessibility. Details of dissected layers are summarized in Table 1 and on the BraDiPho website (<https://bradipho.eu/>).

Photogrammetry acquisition & processing

The photogrammetric setup includes two Sony A7RIII cameras equipped with Voigtlander 65 mm F2 APO-Lanthar lenses, supported by two tripods and macro sledges, a lightbox, a Vivat Turn Table D-26, and a round plate featuring markers and three rulers. The turntable, housed within the lightbox, securely holds the plate. The cameras, positioned at 40 and 60-degree angles, are mounted on the tripods via the macro sledges. A motorized turntable, set to rotate 3 degrees clockwise, connects to both cameras through a splitter. The acquisition process is automated, with the turntable capturing 120 photos per orientation during each full 360-degree rotation. This rotation sequence is repeated twice, with adjustments made to the macro sledges, ensuring an extended depth of field. Each dissection epoch yields 480 photos, all captured at full 42-megapixel resolution, with a shutter speed of 1/60 and aperture of f/11.

The photogrammetric reconstruction of the dissection scene into 3D digital models was carried out iteratively and independently for each epoch using Metashape Pro v1.8. The initial processing involves photo alignment, leveraging markers coded on the rotating plate to compute the orientation and position of each camera during image capture. The three rulers positioned on the plate define the scale of the photogrammetric model. A 3D dense point cloud is generated, achieving an average linear resolution of approximately 0.05 mm. This point cloud serves as the basis for generating a surface mesh model, while high-resolution images are employed to produce photographic texture.

Iterative registration of dissection epochs

The photogrammetric models of each epoch were registered to the radiological space of the specimen using the Iterative Closest Point (ICP) method, integrated into the open-source software CloudCompare v2.11.1 (<https://www.cloudcompare.org/>). The ICP algorithm estimates an affine transformation between surface models under the assumption of preserved morphology across epochs, aside from localized deformations due to tissue removal. The first epoch (*i.e.*, the intact convexity) is registered to the surface model derived from the *ex vivo* T1-MRI. Each subsequent epoch is aligned to the preceding one, namely the one with the closest geometric resemblance, and the cumulative affine transformations from the registration of the previous epochs were applied to preserve consistency across the entire sequence. This iterative approach is implemented for each epoch, effectively mitigating the effects of gradual tissue removal and the related tissue relaxation on alignment accuracy. To assess registration fidelity, we computed the absolute distance between mesh models of consecutive epochs and rendered it as a scalar field on the photogrammetric textures. As shown in Fig. 6A, B, second column for each specimen, minimal displacements localize to preserved regions, while larger distances coincide with zones where tissue has been removed, supporting the reliability of the registration pipeline.

Integration of tractography datasets in each BraDiPho specimen space

The T1-MRI volume of each specimen acquired before dissection provides a reference space for the photogrammetric dissection models of the different epochs. It further introduces the possibility of integrating data from any other radiological space. To integrate the tractography dataset with the layered 3D digital dissection models of each hemisphere, three main transformations are applied:

1. A rigid transformation aligns the photogrammetric model of the first dissection epoch with the *ex vivo* T1-MRI space. This is achieved using the Iterative Closest Point (ICP) method as detailed in the paragraph above.
2. A non-linear transformation maps the MNI ICBM152 2009a T1 template to the *ex vivo* T1-MRI space.
3. The tractography data from the subject's *in vivo* space is mapped to the MNI T1 template space.

These last two transformations are performed with Advanced Normalization Tools (ANTS), specifically the SyN (Symmetric Normalization) technique³⁹, which combines rigid, affine, and diffeomorphic registrations.

The BraDiPho website offers an interactive online service (<https://bradipho.eu/5-tool-convert.html>) that supports the integration of tractography bundles into the dissection models of each specimen. To use this feature, users must upload two files: a tractography file in either *.trk* or *.tck* format and the corresponding T1-MRI nifti file. Any personal information must be removed from these files before uploading, although data will be automatically deleted post-processing for privacy reasons. Once the files are uploaded, the system processes the registration of the user's tractography bundle into the space of the dissection models. The process takes few minutes, after which results are made available for download. The output is provided in two formats: *.ply*, which can be used for visualization in the software CloudCompare, and *.vtk*, which represents the streamlines as polylines for further analysis or visualization.

Tutorial 3 (<https://youtu.be/d-LbJyGmQAQ>) explains how users can explore their own tractography data conjointly with the dissection models. Sample *in vivo* tractography data used in this tutorial represent the structural connectivity of the human AG¹¹ and were sourced from dMRI and T1-MRI images of a single participant of the BIL&GIN

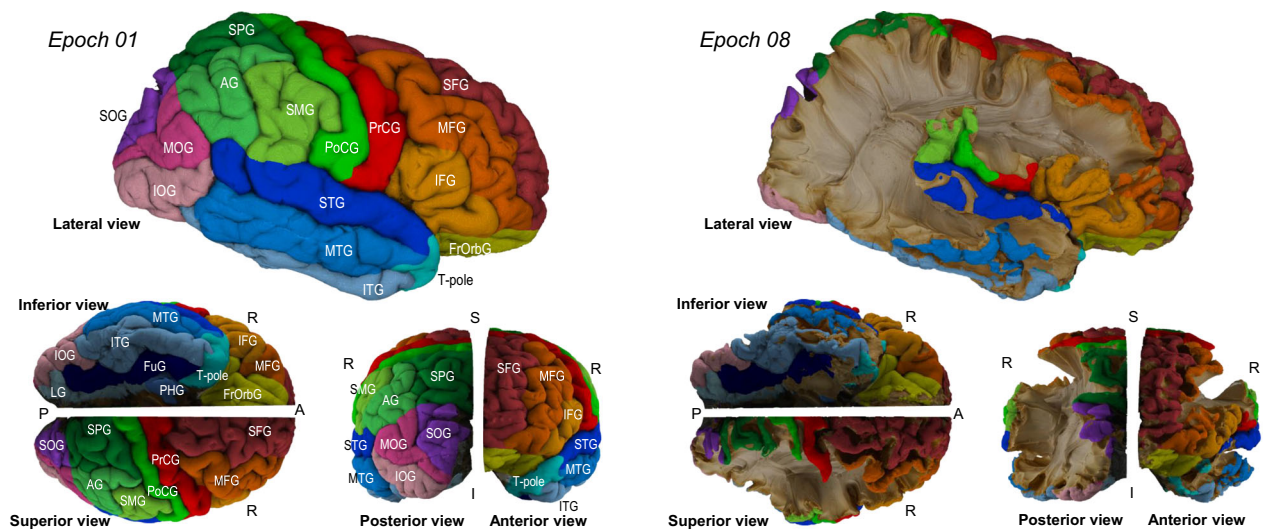
A. Detailed cortical annotations of specimen 01**B. Cortical annotations of the other models**

Fig. 7 | Gyral manual annotations provided for each BraDiPho specimen. Panel (A) reports a detailed illustration of these annotations on spc-01, both on the intact convexity (epoch 01) and as the dissection proceeds (epoch 08). Panel (B) shows cortical annotations on the intact convexity of all the other specimens included in BraDiPho (spc-02, spc-04, spc-10, spc-16, spc-17, spc-18, spc-19). Cortical annotation names and color codes are available in Supplementary Data 1. AG angular gyrus,

FrOrbG fronto-orbital gyrus, FuG fusiform gyrus, IFG inferior frontal gyrus, IOG inferior occipital gyrus, ITG inferior temporal gyrus, LG lingual gyrus, MFG middle frontal gyrus, MOG middle occipital gyrus, MTG middle temporal gyrus, PHG parahippocampal gyrus, PoCG postcentral gyrus, PrCG precentral gyrus, SFG superior frontal gyrus, SMG supramarginal gyrus, SPG superior parietal gyrus, SOG superior occipital gyrus, spc specimen, STG superior temporal gyrus, T-pole temporal pole.

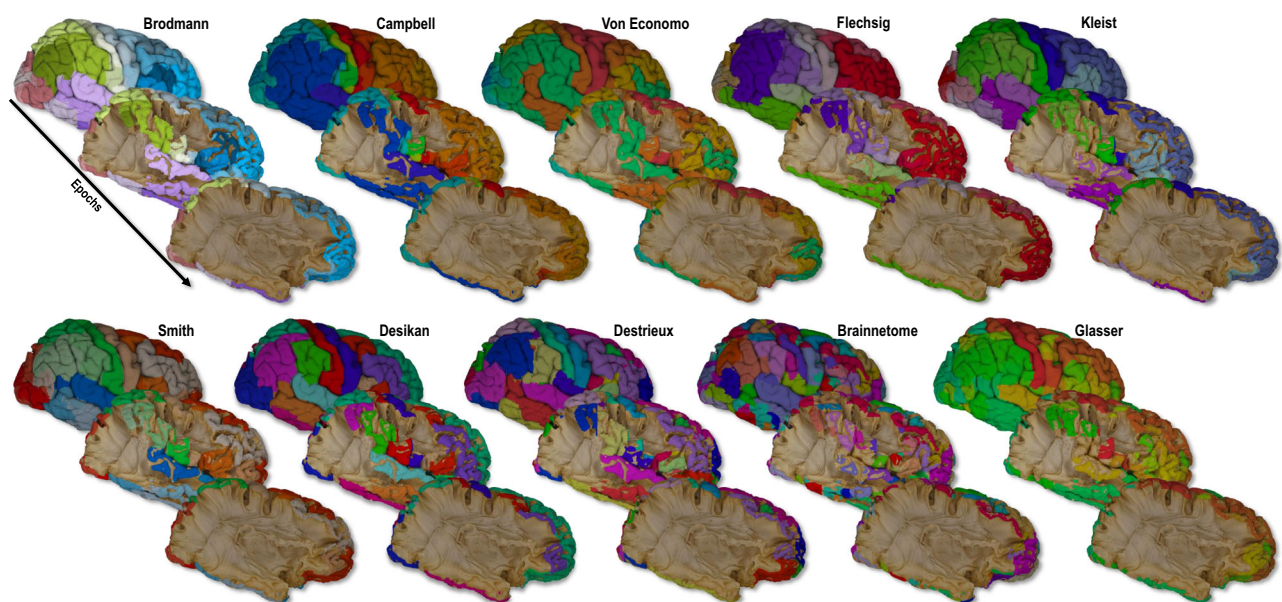


Fig. 8 | Cortical parcellation schemes provided for each BraDiPho specimen. They include classical cyto- and myelo-architectonic atlases (i.e., Brodmann, Campbell, Von Economo, Flechsig, Kleist, Smith) and contemporary connectivity-

based parcellations (i.e., Glasser, Brainnetome, Desikan, Destrieux) projected along all the epochs of each specimen included in BraDiPho. Cortical annotation names and color codes are available in Supplementary Data 1.

database⁴⁰. To prepare these data, probabilistic particle filtering tractography was used to generate a whole-brain tractogram⁴¹. ExTractor_flow was used to extract streamlines of the different short- and long-range bundles, revealing the AG structural connectivity⁴¹.

Derivative data made available for each BraDiPho specimen

All surfaces or point cloud manipulations applied to the dissection models and the derivative datasets included in BraDiPho (i.e., tractography data, manual annotations and cortical parcellation atlases

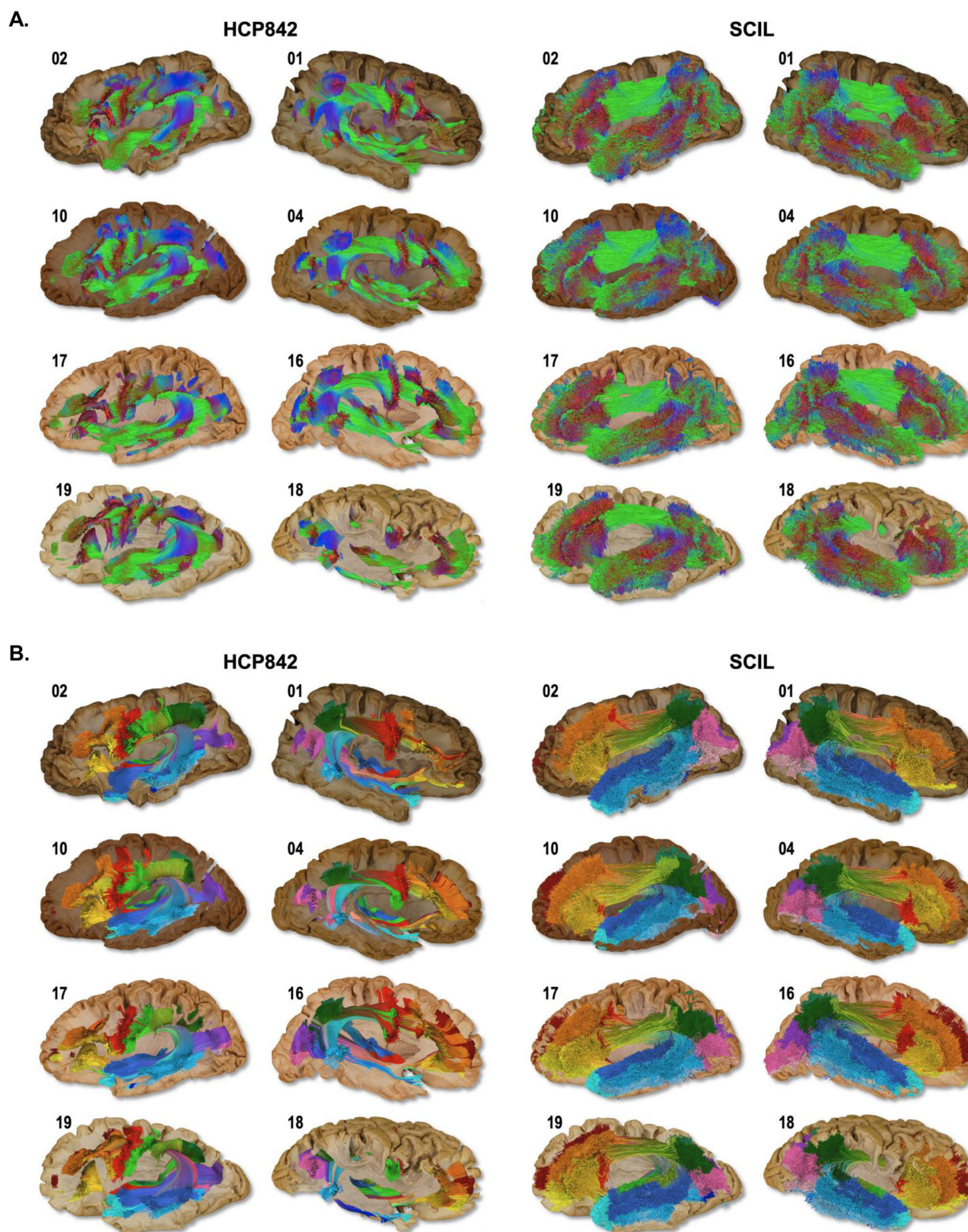


Fig. 9 | Association bundles (AF, SLF, IFOF, UF, ILF, MdLF) of the HCP842 and SCIL atlases integrated in each BraDiPho specimen. A Orientation-based coloring. **B** Cortical termination-based coloring. Each

streamline is color-coded by the cortical annotation its termination belongs to. Color details are available in Supplementary Data 1.

contextualized in the space of the dissection models) were made with a series of Python scripts included in *bradiphopy* (<https://github.com/minilabus/bradiphopy>)²⁸. Volumes (.nii) and tractograms (.trk) manipulation were made with *scilpy* (<https://github.com/scilus/scilpy>), the Sherbrooke Connectivity Imaging Lab (SCIL) Python dMRI processing toolbox.

Cortical surface annotations along all the dissection epochs. To facilitate users of BraDiPho in exploring the anatomical features of each specimen, the cortical surface of the intact convexity has been annotated manually following a thorough sulcal/gyral subdivision in all eight specimens currently available (Fig. 7). Nineteen cortical annotations delineating the gyri of the lateral and basal surfaces are provided. This process constitutes a long and meticulous step in data preparation. Annotations were generated in CloudCompare following the methodology outlined in Tutorial 2 (https://youtu.be/wCO_sbSukMY). In brief, a point cloud made of 100 million points is generated from each mesh model representing the intact convexity of each specimen. Annotations are done by manually outlining the region of interest through successive point picking using the “Segment” tool in CloudCompare. This process is repeated for each “ex-novo” annotation.

Annotations are then transferred from the first epoch of each specimen to all the subsequent epochs by matching neighboring points. The *bdp_match_neighbors.py* script matches the manually drawn annotations of the first epoch with the sampled points in subsequent epochs. The script is run iteratively for each epoch. Finally, the *bdp_colorize_vtk_formats.py* script creates a colorized version of each cortical annotation, useful for illustrative purposes (Fig. 7).

Additional brain parcellation. BraDiPho also includes 12 referenced cortical parcellation atlases registered into the T1-MRI space of each specimen and projected onto the cortical surface of the intact convexity. It comprises the seminal myelo- and cytoarchitectonic microstructural cortical atlases of Campbell (1905), Smith (1907), Brodmann (1909), Flechsig (1920), Von Economo (1925), and Kleist (1934), recently made available as digital reconstruction⁴². We also included the cortical parcellation of the Desikan, Destrieux, Glasser, and Brainnetome atlases^{14,20–22}. A multi-step process is required to map each atlas on the BraDiPho specimens. To ensure accurate registration, the MNI template hosting the original atlases is masked to exclude the cerebellum, matching the T1-MRI of the specimens included in BraDiPho. To improve the accuracy of the cortical surface reconstruction from Freesurfer, we use *mri_synthsr* to generate a (fake) T1-weighted contrast image from the ex vivo image. This helps to enhance the contrast between gray matter and WM, which is crucial for accurate surface reconstruction via *recon-all*. The Freesurfer surface reconstruction is a direct mapping of the specimens’ cortical surface. Registered and parcellated surfaces are saved in the appropriate frame of reference using standard formats for CloudCompare (.ply) and MI-Brain (.nii.gz).

The cortical labels of each atlas parcellation are then transferred along the subsequent dissection epochs as mentioned above for the manual cortical annotations (Fig. 8).

Relevant tractography data registered in the radiological space of each BraDiPho specimen. BraDiPho also includes a selection of representative association bundles of two population-averaged tractography atlases, the HCP842 (https://figshare.com/articles/dataset/Advanced_Atlas_of_80_Bundles_in_MNI_space/7375883)²⁵ and SCIL (<https://zenodo.org/records/10103446>)²⁶ atlases (Fig. 9). This selection includes the AF, the acoustic radiation, the cingulum, the frontal aslant tract, the IFOF, the inferior longitudinal fasciculus, the middle longitudinal fasciculus, the optic radiation, the pyramidal tract, the superior longitudinal fasciculus, the uncinate fasciculus and the vertical occipital fasciculus. These bundles, originally available in the MNI space, are registered to each BraDiPho specimen using both linear and

non-linear transformation computed between the MNI-ICBM152-2009a T1 template and the ex vivo T1-MRI of each specimen.

We are providing tools to co-register files based on their file formats (images, labels, streamlines, etc.) from native space to MNI space and then to the specimen T1-MRI space.

- **Precomputed registrations:** The script uses precomputed registrations from MNI space to each BraDiPho specimen. These registrations are available at <https://zenodo.org/records/11192915>²⁷. This avoids the need to perform time-consuming registration steps for each specimen individually.
- **Code and docker availability:** The code for launching the registration process is available at https://github.com/minilabus/bdp_registration_utils⁴³. This code can be used to reproduce the registration steps and apply the precomputed registrations to other BraDiPho specimens.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The dataset described in this study, including photogrammetric models of ex vivo dissection, tractography atlases, cortical parcellations, and annotations, are openly available at <https://bradiphophy.eu/>.

Code availability

Custom code supporting the integration and rendering of in vivo tractography data into the photogrammetric models are available at <https://zenodo.org/records/11192915>²⁷ and https://github.com/minilabus/bdp_registration_utils⁴³, respectively, and they are implemented as an online service at <https://bradiphophy.eu/5-tool-convert.html>. All codes developed for supporting conversions between volumes and surfaces, as well as across different file formats, are also available at <https://github.com/minilabus/bradiphophy/tree/main/scripts>²⁸.

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Competing interests

The authors declare no competing interests.

Additional information

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